Course Project

Cyber-Physical Operations on a 5-Bus System: Power Flow, OPF, N-1 Contingency Analysis, and Security-Constrained Re-Dispatch

Course	Cyber-Physical Systems for Power Processing and Smart Grids - AGH 2026
Instructor	Charalambos (Harrys) Konstantinou
Weight	50% of final grade
Released	May 2026
Due	31 May 2026, 23:59 CET (firm)

1 Overview

This project asks you to build a small, end-to-end software pipeline that ties together every operational function we have studied in Lectures 1–4: Y_{bus} and DC power flow, the PTDF/LODF sensitivity framework, DC-OPF and locational marginal prices, full N-1 contingency screening, and security-constrained re-dispatch (SCOPF). The cyber-physical lens is the point: contingencies are not only random physical failures - they can also be the deliberate goal of an adversary who trips a line, induces a relay misoperation, or otherwise removes a network element. The decisions an operator makes *before* a contingency happens (do I dispatch cheaply, or do I dispatch securely?) directly determine the system's resilience to both classes of events. You will quantify that trade-off in dollars per hour.

2 Learning objectives

- Build a self-contained DC-power-system toolbox in any tool of your choice (Python, MATLAB, Julia, AMPL, GAMS) that solves Y_{bus}, DC power flow, PTDF, and DCOPF.
- Implement the Line Outage Distribution Factor (LODF) matrix and use it to perform a full N-1 contingency screening.
- Formulate and solve a Security-Constrained DCOPF (SCOPF) that respects both base-case and post-contingency line ratings for one chosen contingency.
- Quantify the cost of security (ΔC) and place a base-DCOPF dispatch and an SCOPF dispatch into the four operating-state framework of Lecture 4 (slides 80–85).
- Communicate technical results in a clear written report.

3 System under study - a custom 5-bus network

Per-unit on a 100 MVA base. Lossless DC model. **Bus 4 is the slack** with $\theta_4 = 0$, $|V_4| = 1.00$ p.u.

Bus data

Bus	Type	$P_g^{\min}$ (MW)	$P_g^{\max}$ (MW)	Cost (\$/MWh)	Load (MW)
1	PV	20	180	12.40	0
2	PQ	–	–	–	90
3	PV	0	100	22.60	0
4	Slack	0	200	31.10	0
5	PQ	–	–	–	60

Branch data

Line	From	To	X (p.u.)	Rating (MW)
L_1	1	2	0.094	100
L_2	1	4	0.135	100
L_3	2	3	0.181	70
L_4	3	4	0.122	90
L_5	4	5	0.107	80
L_6	2	5	0.165	60

Reproducibility. Print the seed value `31052026` (the project deadline date) once at the top of your run banner. If any randomness appears in your code (e.g., for tie-breaking or starting points), seed it with this value. We will run your code and check that key quantities match.

4 Required code structure

Your repository must contain a single module/file (e.g., `cps_grid.py`, `cps_grid.m`, `cps_grid.jl`, `cps_grid.mod`, etc.) defining *at least* these procedures, with the input/output specification below (use the equivalent calling convention of your tool). You may add more as needed:

```
build_B(branches, n_buses) # B' susceptance matrix
solve_dc_pf(B, P_inj_pu, slack_bus_1idx)
build_PTDF(branches, n_buses, slack_bus_1idx)
build_LODF(branches, n_buses, slack_bus_1idx) # returns L_LL matrix
solve_dcopf(buses, branches, gens, slack_bus_1idx)
run_n1_screening(branches, PTDF, LODF, P_inj_pu, ratings_MW)
    # returns one record per outaged line, with fields
    # outage, post_flows, max_loading_pct, worst_line
solve_scopf(buses, branches, gens, slack_bus_1idx, contingencies)
    # contingencies:  list of line indices to enforce N-1 against
```

You may use any standard linear-algebra and LP/MILP solver capability of your tool, including but not limited to: `numpy/scipy`, Pyomo or CVXPY in Python; the backslash operator, `linprog`, or YALMIP in MATLAB; `LinearAlgebra` and JuMP in Julia; AMPL; GAMS. Any LP solver is fine (HiGHS, CBC, GLPK, CPLEX, Gurobi, etc.). **You may not use any high-level power-systems library** (e.g., MATPOWER, PYPOWER, PandaPower, PowerFactory, OpenDSS, PowerWorld, GridCal).

Phase 1. Power-system foundations (15 points)

(a) Implement `build_B` and `solve_dc_pf`. Solve the base-case DC power flow (with all generators dispatched proportionally to fill demand; you may pick any feasible split, e.g., $G_1 = 110$, $G_3 = 40$, $G_4 = 0$). Report the angle vector and the six line flows in a table.

(b) Implement `build_PTDF` (slack at bus 4). Report the full 6×5 PTDF matrix to four decimal places. Confirm numerically that the column for bus 4 is identically zero.

(c) Cross-check: compute the line flows of part (a) two different ways: via direct `solve_dc_pf` and via `PTDF · P_inj`. The two answers must match to within 10^{-6} on every line. Show the comparison.

Phase 2. DCOPF and locational marginal prices (15 points)

(a) Implement `solve_dcopf` using any LP solver of your choice. Solve the base case (all line ratings as in the branch data table). Report: optimal dispatch, line flows, total cost ($/h), and LMP at every bus to two decimal places.

(b) Identify any binding line constraints. Are all five LMPs equal?

(c) **Engineer congestion:** reduce the rating of L_2 from 100 MW to 30 MW. Re-solve. Report the new dispatch, LMPs, and the dual variable μ_{L_2} on the binding line constraint. Comment on which buses are now "expensive" and which are "cheap".

Phase 3. N-1 contingency screening with LODF (20 points)

The Line Outage Distribution Factor (LODF) matrix $L \in \mathbb{R}^{n_\ell \times n_\ell}$ tells you, for every monitored line m and every outaged line k, the fractional change in m's flow when k is removed:

$$\Delta P_m^{(k)} \;=\; L_{m,k}\, P_k^{(0)},$$

where $P_k^{(0)}$ is the pre-contingency flow on line k. The post-contingency flow on m is then $P_m^{(0)} + \Delta P_m^{(k)}$. Standard derivation (Wood-Wollenberg, ch. 11): given the PTDF matrix Φ, $L_{m,k} = \Phi_{m,k}/(1 - \Phi_{k,k})$ for $m \neq k$, with $L_{k,k} = -1$ by convention. Here $\Phi_{m,k} \equiv \Phi_{m,f_k} - \Phi_{m,t_k}$ is the bus-to-bus PTDF for a unit transaction along line k (i.e., +1 MW injected at the from-bus of line k, withdrawn at its to-bus), the same notation used in HW3.

(a) Implement `build_LODF`. Report the full 6×6 LODF matrix to four decimal places. Report explicitly the value of L_{L_3,L_1} (the impact on L_3 of outaging L_1) and confirm $L_{k,k} = -1$ for every k.

(b) Implement `run_n1_screening`. Using the dispatch from your *base* DCOPF (Phase 2(a), with all line ratings as in the branch-data table), for each of the six possible single-line outages, compute the post-contingency flows on the remaining lines via the LODF formula. Build a contingency table with one row per outage, listing: outaged line; max post-contingency loading (in % of rating, computed over the surviving lines); the surviving line that is most loaded; whether any line exceeds its rating.

(c) **Cross-check against full re-solve.** Pick the contingency that the LODF screening identifies as the worst (largest max post-contingency loading). For that single outage, recompute the post-contingency flows by directly removing the outaged line from the network, rebuilding B', and re-running `solve_dc_pf`. The two answers (LODF-predicted and directly re-solved) must agree to within 10^{-6} on every surviving line. Report the comparison.

(d) In one or two sentences, name a plausible physical *or* adversarial event that could cause this worst contingency in practice (lightning fault, breaker mis-operation, relay misconfiguration triggered by an attacker, etc.). This is a sentence, not an essay.

Phase 4. Security-constrained DCOPF (25 points)

A preventive SCOPF chooses a base-case dispatch that satisfies, simultaneously: (i) the base-case power balance and line ratings, and (ii) the post-contingency line ratings for every contingency in a chosen list. For a single contingency k, the post-contingency flow on line m as a function of injections is

$$P_m^{(k)}(P) \;=\; \Phi_{m,:}P \;+\; L_{m,k}\,\Phi_{k,:}P,$$

i.e., a linear function of the injection vector P. Adding the post-contingency rating constraints $|P_m^{(k)}(P)| \leq P_m^{\max}$ for every m keeps the SCOPF a single LP.

(a) Implement `solve_scopf` that takes a list of contingency indices and adds the corresponding post-contingency line-rating constraints to the DCOPF formulation. Solve it for the *single worst* contingency identified in Phase 3(c). Report: the SCOPF dispatch, the SCOPF base-case line flows, the SCOPF total cost ($/h), and the SCOPF LMPs at every bus to two decimal places.

(b) Compare side-by-side: list the base DCOPF dispatch (from Phase 2(a)) and the SCOPF dispatch in a single table. Identify which generator(s) had to be moved off their economic optimum, and by how much.

(c) **Cost of security.** Compute $\Delta C = C_{\text{SCOPF}} - C_{\text{DCOPF}}$ in $/h, and report it both in absolute terms and as a percentage of the base DCOPF cost. This is exactly the price paid (per hour, in this dispatch interval) to remain secure against the chosen contingency.

(d) **Verify.** Apply the worst contingency from Phase 3 to the SCOPF dispatch (use LODF to compute the post-contingency flows under the SCOPF injection vector). Confirm that no surviving line exceeds its rating. Report the post-contingency loading for every surviving line under the SCOPF dispatch.

Phase 5. Operating-state analysis and reflection (15 points)

Lecture 4 (slides 80–85) introduced four operating states: *normal-state dispatch*, *post-contingency*, *secure dispatch*, and *secure post-contingency*. This phase places your two dispatches (base DCOPF and the single-contingency SCOPF from Phase 4(a)) into that framework and quantifies the trade-off.

(a) Build the following 2×2 table. Rows = {Base DCOPF, SCOPF (single worst contingency)}. Columns = {pre-contingency, post-contingency under the worst contingency}. Each cell reports two numbers: max line loading (% of rating, over all lines that exist in that scenario) and total cost ($/h). The four cells correspond exactly to the four operating states from slide 81.

(b) In one paragraph (max 150 words), interpret the table. Quote your numbers. Make explicit the trade-off: SCOPF pays ΔC $/h all the time, in exchange for not being exposed to a specific overload if the contingency occurs.

(c) **Cyber-physical reflection (100–150 words).** An operator must decide before the dispatch interval whether to run base DCOPF or SCOPF. Discuss the decision when contingencies can be *adversarial* as well as random: an attacker can deliberately trigger a line trip (relay manipulation, breaker mis-operation, coordinated physical-cyber action). Reference at least one published incident in the power sector (e.g., Ukraine 2015/2016, Aurora 2007, or any documented advanced persistent threat campaign on grid operators). Two to three sentences on the incident, then your point about the dispatch decision.

Phase 6. Report and submission (10 points)

Submit a single `.zip` archive named `Surname_Name_Project.zip` containing:

- **report.pdf** (max **8 pages**, single column, 11 pt). Include: a short introduction, one section per phase with results (tables, plots), the contingency table from Phase 3(b), the bar chart from Phase 5(c), the cyber-physical reflection from Phase 5(d), and a brief conclusion.
- e.g., **cps_grid.py**, i.e., the module with all required functions.
- e.g., **run_project.py**, a single script that, when run end-to-end, reproduces every numerical answer and every plot in your report. The script must print, at the start of its output:

  - your single-line author/timestamp banner;
  - the current tool version, e.g., (`sys.version`), `numpy` version, solver name, and solver version;
  - the seed used (`31052026`).

- **README.md**: environment setup and how to run.

We will run your (e.g., `run_project.py`) on a clean machine with a fixed environment. Numerical answers in your report that disagree with what your code prints will be graded as wrong.

5 Grading rubric

Component	Points
Phase 1, Y_{bus}, DC PF, PTDF	15
Phase 2, DCOPF and LMPs	15
Phase 3, LODF and N-1 contingency screening	20
Phase 4, Security-constrained DCOPF	25
Phase 5, Operating-state analysis and reflection	15
Phase 6, Report quality, code clarity, reproducibility	10
Total	**100**

6 Suggested timeline

Day	Milestone
1	Phases 1 and 2 done: Y_{bus}, DC PF, PTDF, DCOPF, LMPs (with congestion).
2	Phase 3 (LODF, N-1 screening, cross-check) and Phase 4 (single-contingency SCOPF, cost of security).
3	Phase 5 (2×2 table, bar chart, reflection) and Phase 6 (report, plots, polishing, code clean-up).

7 Recommended reading

- A. Wood, B. Wollenberg, G. Sheblé, *Power Generation, Operation, and Control*, 3rd ed., Wiley, 2014. Ch. 6 (DCOPF, LMPs), Ch. 11 (contingency analysis, LODF, SCOPF).
- B. Stott, J. Jardim, O. Alsaç, "DC Power Flow Revisited," *IEEE Trans. Power Systems*, vol. 24, no. 3, pp. 1290–1300, 2009.
- F. Capitanescu *et al.*, "State-of-the-art, challenges, and future trends in security-constrained optimal power flow," *Electric Power Systems Research*, vol. 81, no. 8, pp. 1731–1741, 2011.
- For the cyber-physical reflection in Phase 5: SANS, "Analysis of the Cyber Attack on the Ukrainian Power Grid," E-ISAC, 2016 (the Ukraine 2015 incident); D. U. Case, "Analysis of the cyber attack on the Ukrainian power grid" (Ukraine 2016, Industroyer/CrashOverride).

8 Submission and policy

Late submissions are penalized 10% per day, up to three days; after that, no credit. The project must be completed individually unless explicitly approved as a pair. **If you use generative AI tools, disclose this on the first page of your report.** You are responsible for verifying every numerical answer against your own code; AI-fabricated numbers that do not match what your code does (i.e., `run_project.py`) prints will be graded as wrong.